Hyeokjoon Kwon

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Research Interests

Multimodal Representation Learning; SLAM / Spatial AI

Education

Carnegie Mellon University Pittsburgh, USA
M.S. in Robotics · GPA 4.17/4.0 *Sep 2025 – Present*

- Relevant Coursework: Advanced Computer Vision (16-820), Localization and Mapping (16-833), Introduction to Robot Learning (16-831)

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
B.S./M.S. in Aerospace Engineering, GPA 3.9/4.3 (B.S., GPA 3.73/4.3) *Feb 2021*

- M.S. Thesis: *Low-thrust Spacecraft Trajectory and Guidance Law Design Using Reinforcement Learning*

Research Experience

AirLab, Carnegie Mellon University Pittsburgh, USA
Graduate Research Assistant (Supervisors: Sebastian Scherer, Wenshan Wang) *Sep 2025 – Present*

- **Multimodal Representation Learning for Dense Correspondences.** Designed model architectures and training recipes to extend RGB-based universal flow models (e.g., UFM) to multimodal and cross-modal settings spanning event, thermal, and LiDAR sensors; curated simulated and real datasets with pixel-level alignment across heterogeneous sensors for large-scale training.
- **Target Re-Identification and Localization for Search & Rescue (Triage).** Designing an end-to-end system to detect, localize, and re-identify casualties for rapid triage support to first responders, leveraging monocular RGB SLAM algorithms together with external sensors; applying contrastive learning to extract distinguishable features for detected targets.

Aerospace Systems & Control Lab, KAIST Daejeon, Korea
Research Assistant (Supervisor: Hyochoong Bang) *Feb 2019 – Feb 2021*

- **Reinforcement Learning for Orbit Transfer.** Led research on low-thrust spaceflight guidance for continuous orbit transfer using reinforcement learning, addressing key limitations of traditional trajectory optimization such as high computational cost and sensitivity to initial guesses.
- **CubeSat for Earth Observation.** Performed communication link budget analysis for a 3U CubeSat (RANDEV), defined mission objectives and engineering requirements, and prepared hardware components.
- **Sensor Fusion for Pose Estimation.** Designed a loosely coupled GNSS/IMU Extended Kalman Filter for initial pose estimation of a wing-mounted radar, integrating dynamics of flexible wings, body, and radar.

Work Experience (Selected)

The Boeing Company Seoul, Korea
Machine Learning Researcher *Mar 2024 – Jul 2025*

- **1M Parts Recognition.** Developed a classification algorithm for over 1 million unique categories of aerospace parts (fasteners, bearings, etc.) using metric and representation learning; planned for deployment to Boeing Distribution Services.
- **Wing Hole Inspection.** Developed a homography estimation algorithm using learning-based feature matching with a quantitative evaluation method to track hole and fastener locations on B-777X wings, ensuring accurate placement.

- **Composites and Sealant Inspection.** Designed a data collection process at B-787 wing factories and developed an anomaly detection algorithm to identify foreign object debris on composites and air entrapment in sealants; implemented prompt-based video segmentation to accelerate annotation.

Nearthlab Inc. Seoul, Korea
Machine Learning Engineer *Jan 2022 – Mar 2024*

- **Perception for Autonomous Drone Flight.** Developed and deployed a rotated object detection algorithm for autonomous inspection on Android and iOS using TFLite and CoreML; defined metrics, conducted error analysis, and enriched datasets across adverse weather, views, and lighting.
- **Precision Seeding with Drone Imagery and 3D Reconstruction.** Developed an algorithm to generate precise GPS coordinates for tractor seeding using drone imagery; combined prompt-based segmentation with 3D reconstruction to achieve geodetic accuracy within 10 cm.
- **High-Resolution Image Stitching for Wind Turbines.** Developed a 48 MP image stitching algorithm fusing deep feature matching with GPS/INS/LiDAR for textureless wind turbine blade imagery; accelerated inspection by 7× and provided precise 3D locations of defects. Placed 7th of 642 teams (Top 1%) in the CVPR 2022 Image Matching Challenge by Google Research (writeup).
- **ML-based Defect Detection.** Led product development for wind turbine defect detection, including data definition and annotation, training of segmentation and object detection models, and deployment on MySQL and Kubernetes; optimized preprocessing and inference pipelines with multiprocessing, batching, and ONNX.

Nearthlab Inc. Seoul, Korea
Aerospace Engineer *Feb 2021 – Jan 2022*

- **Flight Software for Autonomous Drone Inspection.** Developed flight software enabling drones to autonomously track and photograph wind turbine blades while safely maintaining distance to satisfy resolution requirements; applied OOP and design patterns for real-time onboard implementation in C++.
- **SITL System and 6DoF Flight Simulator.** Developed a Simulation-In-The-Loop system with a 6DoF flight simulator using Gazebo, ROS, and C++; implemented efficient numerical integration based on rigid-body dynamics and enhanced fidelity with second-order motor dynamics.

Publications

- **(Under Review) H. Kwon**[*], J. Cai[*], R. Li[*], K. Bhandari, G. Hemkumar, P. Maheshwari, Y. Qiu, Y. Zhang, S. Scherer, W. Wang (2026). "TartanMatch: Towards Universal Dense Correspondence Across Modalities." *arXiv preprint arXiv:2610.14513*. [arXiv] [Website] [Code]
- **(Conference Proceedings) H. Kwon**, S. Oghim, H. Bang (2022). "Autonomous Guidance for Multi-revolution Low-thrust Orbit Transfer via Reinforcement Learning." *Proceedings of the AAS/AIAA Space Flight Mechanics Meeting*, AAS 21-315, pp. 783–798. [PDF]
- **(Peer-reviewed Article)** D. Y. Lee, **H. Kwon**, D. Lee, H. Lee, Y. Jung, J. Jeong, S. Park, S. Lee, J. H. Park, M. Tahk, H. Bang, J. Ahn (2021). "Performance Analysis of an Integrated Navigation of an Airborne AESA Radar." *Journal of the Korean Society for Aeronautical & Space Sciences*, 49(4), pp. 281–290. [link]

Patents

- D. Shin, S. Baek, **H. Kwon** (2026). "A method and an apparatus for merging image based on contour of object." KR1029469660000.
- **H. Kwon**, S. Baek, D. Shin, Y. Choi (2025). "A method and an apparatus for displaying defects of an object in an image." KR1027941550000.
- **H. Kwon**, S. Baek, D. Shin (2025). "A method and an apparatus for merging images." KR1027575010000.
- J. Moon, S. Baek, **H. Kwon** (2022). "AI apparatus and method for image stitching for inspecting a wind turbine." KR102467590B9, EP4328850A3, US20240070850A1.